

TraceMail AI

Email Forensic Investigation Report

Case ID: TM-AE5A25BB
Report Generated: 2026-09-06T08:07:53.474494

1. Executive Summary

Severity	HIGH
Risk Score	70/100
Threat Type	Business Email Compromise / Financial Phishing

Assessment: The email contains multiple suspicious indicators and requires analyst review before any user action.

2. Email Evidence

Filename	test-phishing-002.eml
Sender	accounts-payable@northstar-finance.example
Recipient	employee@company.example
Reply-To	payment-review@northstar-secure.example
Subject	Urgent Invoice Payment Verification Required
Date	Fri, 28 Aug 2026 22:25:00 +0530
Message ID	<test-phishing-002@northstar-finance.example>

3. Email Authentication

Authentication Check	Result
SPF	
DKIM Signature	Missing
Authentication Summary	

These checks help determine whether the sender's domain was properly authenticated. Failed checks increase the suspicion level but should be considered together with the other evidence.

4. Email Header & Relay Analysis

Total Relay Hops: 0
Earliest Observed Public IP: N/A

Note: The earliest observed IP represents source infrastructure visible in the email headers. It should not automatically be treated as the attacker's physical location.

5. IP Intelligence

No public IP intelligence was available.

6. URL & Domain Intelligence

URLs Found: 1

- <https://invoice-verification.northstar-secure.example/verify?id=83921>

Domains Found: invoice-verification.northstar-secure.example

The domain and URL analysis checks for suspicious naming patterns, unsafe links and indicators that may suggest impersonation or phishing activity.

7. Threat DNA

Threat DNA is a compact summary of the email's important security characteristics. It helps investigators quickly compare this email with other suspicious emails.

Security Feature	Finding
SPF Authentication	UNKNOWN
DKIM Authentication	MISSING
DMARC Authentication	UNKNOWN
Contains URL	Yes
Number of URLs	1
Financial / Payment Intent	Yes
Urgency Language	Yes
Reply-To Different From Sender	Yes
Contains Attachment	No
Email Relay Hops	0
Suspicious / Impersonating Domain	No

Threat DNA Fingerprint

SPF-UNKNOWN|DKIM-MISSING|DMARC-UNKNOWN|URL-1|FIN-1|URG-1|RPLY-1|ATT-0|HOP-0|TYPO-0

Detected Pattern: Business Email Compromise / Financial Phishing

8. Identified Risk Factors

The following indicators contributed to the threat assessment.

DKIM authentication is missing or failed [MEDIUM]

Reply-To domain differs from sender domain [HIGH]

Urgency language detected [MEDIUM]

Detected terms: urgent, immediately

Financial/payment-related intent detected [HIGH]

Detected terms: payment, invoice, account

Credential-related language detected [HIGH]

Detected terms: verification

1 URL(s) detected in email [MEDIUM]

9. Evidence Integrity

Integrity Method: SHA-256

Evidence Hash

2eb550fc3fbf794f1ec3cb7ca1e795c03cbe4eca689753b5dbd1b023f10caff4

This SHA-256 hash acts as a digital fingerprint of the analyzed email evidence. If the original evidence changes, its hash will also change, helping investigators detect unexpected modification.

10. Investigator Conclusion

The email contains multiple suspicious indicators and requires analyst review before any user action.

TraceMail AI

Email Threat Detection & Forensic Intelligence Platform